MIL-C-6136A
10 DECEMBER 1965

Superseding
MIL-C-6136
27 April 1950

MILITARY SPECIFICATION

# CONDUIT: ELECTRICAL, FLEXIBLE, SHIELDED, ALUMINUM ALLOY FOR AIRCRAFT INSTALLATIONS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope.- This specification covers the requirements for shielded electrical conduit for aircraft installations.

1.2 Classification.- Flexible shielded conduit shall be furnished in two types, as specified (see 6.2):

Type I - Without outer synthetic compounded covering

Type II - With outer synthetic compounded covering

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| QQ-A-225/2 | Aluminum Alloy Bar, Rod, and Wire; Rolled, Drawn, or Cold Finished, 3003 |
| QQ-A-250/2 | Aluminum Alloy 3003, Plate and Sheet |
| QQ-A-430 | Aluminum Alloy Rod and Wire; for Rivets and Cold Heading |

Military

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of |
| MIL-R-6855 | Rubber, Synthetic, Sheeted, Molded, and Extruded |
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking of Shipments |

FSC 5975

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:39Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Materials.- Materials shall conform to applicable specifications and be as specified herein and on applicable drawings. Materials which are not covered by specifications or which are not specifically described herein shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.1.1 Tubing material.- The tubing shall be formed from aluminum alloy sheet conforming to QQ-A-250/2 or from flattened wire conforming to QQ-A-225/2.

3.1.2 Packing material.- The packing material used shall be such as not to cause corrosion when the tubing is exposed to salt atmosphere. If cotton packing is used it shall be impregnated with bituminous paint or other acceptable compound.

3.1.3 Braid material.- The wire from which the braid is made shall be aluminum alloy conforming to QQ-A-430, temper O.

3.1.4 Outer covering material (type II).- The outer cover of type II conduit shall be of synthetic rubber conforming to class 2 of MIL-R-6855 and shall be of uniform thickness and free from pitting.

3.2 Design and construction.-

3.2.1 Complete conduit.- Type I complete conduit shall consist of flexible aluminum alloy tubing covered with a woven aluminum alloy wire braid (see figure 1). Type II complete conduit shall be identical to type I, except that it shall be provided with an outer protective synthetic rubber covering.

3.2.1.1 Sizes and dimensions.- The nominal sizes and inside and outside diameters of the complete conduit shall be as specified in table I.

3.2.1.2 Bending radius.- All conduit (minus the outer synthetic rubber covering of type II) shall withstand bending in a loop, with the inside radius as specified in table I, without damage.

3.2.2 Tubing.- The tubing shall be of a rugged, full four-wall interlocking type, suitably packed to provide the specified watertightness (see 3.2.2.3). The tubing shall be made by helically coiling a formed aluminum alloy strip in such manner that it will not unravel or show loose ends when cut. The width of the strip and pitch of the helix shall be designed to provide for the required bending radius (see 3.2.1.2).

3.2.2.1 Tubing dimensions.- The dimensions of the tubing and the thickness of the strip used in its forming shall be as specified in table I.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:39Z
Check the source to verify that this is the current version before use.

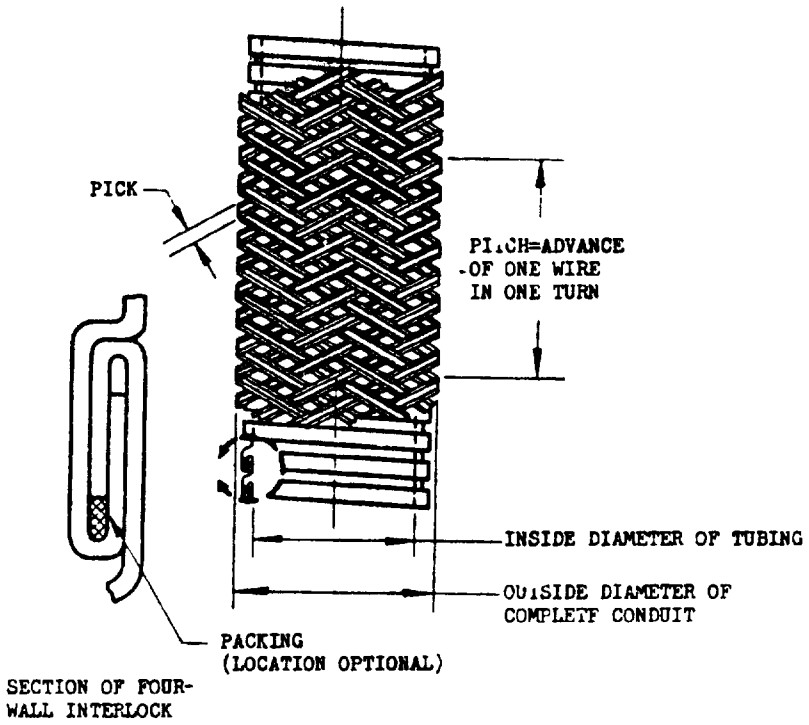



FIGURE 1. Shielded conduit type I and type II (before outer covering is applied)



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:39Z
Check the source to verify that this is the current version before use.

TABLE I. Sizes and dimensions

| Nom. size (ID) | Complete conduit | | | | | | | Tubing | | Shielding braid | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Inside diameter | | Outside diameter type I | | Outside diameter type II | Bending radius (inside) | Load for bending test | Thickness of strip | Yield load | Wires per pick (max.) | | Wire diameter | | Minimum coverage |
| | Inches | | Inches | | Inches | Inches | Lb. | Inch | Lb. | 2/2 weave | 1/1 weave | Inch | Inch | Percent |
| Inches | Min | Max | Min | Max | Max | Max | Max | Nom. | Min | | | Min | Max | |
| 3/16 | 0.183 | 0.193 | 0.318 | 0.338 | 0.386 | 2 1/4 | 1 | 0.010 | 50 | 6 | 8 | 0.0075 | 0.0085 | 85 |
| 1/4 | .245 | .255 | .380 | .400 | .448 | 2 3/4 | 1 | .010 | 75 | 6 | 8 | .0075 | .0085 | 85 |
| 3/8 | .370 | .380 | .505 | .525 | .573 | 3 3/4 | 1 | .010 | 85 | 6 | 8 | .0075 | .0085 | 85 |
| 1/2 | .495 | .505 | .630 | .650 | .698 | 3 3/4 | 2 | .010 | 100 | 6 | 8 | .0075 | .0085 | 85 |
| 5/8 | .620 | .630 | .755 | .775 | .823 | 3 3/4 | 2 | .010 | 105 | 6 | 8 | .0075 | .0085 | 85 |
| 3/4 | .745 | .755 | .910 | .930 | .983 | 4 1/4 | 2 | .010 | 110 | 7 | 14 | .0095 | .0105 | 90 |
| 1 | .995 | 1.005 | 1.200 | 1.220 | 1.345 | 5 3/4 | 2 | .012 | 120 | 7 | 14 | .0095 | .0105 | 90 |
| 1 1/4 | 1.245 | 1.255 | 1.450 | 1.475 | 1.600 | 8 | 2 | .012 | 160 | 7 | 14 | .0095 | .0105 | 90 |
| 1 1/2 | 1.495 | 1.505 | 1.703 | 1.728 | 1.853 | 8 1/4 | 7 1/2 | .012 | 190 | 7 | 15 | .0095 | .0105 | 90 |
| 1 3/4 | 1.745 | 1.755 | 1.953 | 1.978 | 2.103 | 9 | 7 1/2 | .012 | 225 | 8 | 15 | .0095 | .0105 | 90 |
| 2 | 1.995 | 2.005 | 2.208 | 2.233 | 2.353 | 9 3/4 | 7 1/2 | .012 | 265 | 8 | 16 | .0121 | .0131 | 90 |
| 2 1/2 | 2.495 | 2.505 | 2.720 | 2.745 | 2.870 | 10 | 7 1/2 | .012 | 335 | 8 | 16 | .0121 | .0131 | 90 |

TABLE II. Piece lengths

| Nominal size (ID) | Nominal Extended length | Minimum acceptable extended length |
|---|---|---|
| Inches | Feet | Feet |
| 3/16 | 50 | 25 |
| 1/4 | 50 | 20 |
| 3/8 | 50 | 20 |
| 1/2 | 40 | 15 |
| 5/8 | 40 | 14 |
| 3/4 | 40 | 10 |
| 1 | 30 | 10 |
| 1 1/4 | 20 | 10 |
| 1 1/2 | 17 | 10 |
| 1 3/4 | 15 | 10 |
| 2 | 13 | 10 |
| 2 1/2 | 10 | 10 |

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:39Z
Check the source to verify that this is the current version before use.

3.2.2.2 Tubing yield load.- The yield load of the tubing (conduit with outer covering and braid removed) shall be as specified in table I.

3.2.2.3 Water tightness.- Class I tubing from which the shielding braid has been removed, or class II tubing from which the outer covering and the shielding braid have been removed and which has been flexed to the bending radius specified in table I, shall not leak more than one-third of its interior volume per hour when tested as specified in 4.5.2.

3.2.3 Braid.- The braid shall be applied in such manner that the finished conduit may be bent to the specified radius without damage (see 3.2.1.2). The braid shall fit the tubing closely enough to prevent appreciable bunching or undue slipping.

3.2.3.1 Braid design.- The weave of the braid shall be either of the basket types, over two and under two, or over one and under one. The diameter of the wire used for the braid and the number of wires per pick shall be as specified in table I. Each pick shall contain the same number of wires.

3.2.3.2 Braid coverage.- The total number of wires in the complete braid and the number of parallel wires per inch length of conduit shall be such as to provide the percentage of coverage specified in table I, when calculated by the method specified in 4.5.4.

3.2.4 Physical properties of outer cover.- The physical properties of the outer synthetic rubber covering, both before and after testing, shall conform to class 2 of MIL-R-6855.

3.3 Marking.- Types I and II conduit shall be marked by either of the following methods:

(a) Continuous or intermittent lengths of waterproof tape marked with the manufacturer's name or trademark at intervals of not more than 1 foot shall be incorporated between the metal tubing and the braid.

(b) Manufacturer's identification shall be placed at not more than 3-inch intervals on the inner side of the spiral convolutions of the conduit.

3.4 Length of cut.- The maximum permissible length of a single piece within any quantity ordered is limited only by manufacturing and handling facilities. Not less than 85 percent of the total quantity of each size on order shall be furnished in lengths equal to or greater than the nominal lengths specified in table II. Pieces shorter than the minimum acceptable lengths specified in table II will not be accepted. All lengths shall be continuous, unspliced, and shall be measured when the conduit is fully extended under a tension of approximately 10 percent of the yield load specified in table I.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:39Z
Check the source to verify that this is the current version before use.

3.4.1 End protection.- Each end of each length of type I conduit shall be provided with a removable ferrule or with suitable seizing or tape, which will prevent unraveling of the braid and cover the ends of the braid wires.

3.5 Workmanship.- All details of manufacture shall be in accordance with the best practice for aircraft flexible conduit. The conduit shall be free from all defects which would adversely affect its serviceability.

3.5.1 Interior.- The interior surface of the conduit shall be free from burrs or sharp edges which might cause abrasion of cable insulation, and shall be free from obstructions.

3.5.2 Splices.- Splices in braids shall be neatly made, and shall cause no sharp or rought projections on the conduit. There shall be not more than one spliced pick in any 2-foot length of conduit. All loose ends shall trimmed closely.

3.5.3 Ends.- The ends of each length of conduit shall be cut square, and shall be free from burrs or sharp edges which would interfere with the assembly of the conduit with parts with which it is intended to be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.- Unless otherwise specified, sampling for examination and tests shall be in accordance with MIL-STD-105.

4.3 Inspection lot.- Conduit of one type and size presented at one time shall be considered a lot. The inspection lot size shall be determined by the number of coils or reels presented. The sample unit for examination shall consist of 4 consecutive feet of conduit.

4.3.1 Visual examination.- The conduit shall be examined for defects in material, design, construction, marking, and workmanship. Any area of nonconformance with specified requirements shall be classified a defect. The inspection level shall be Level II with an AQL of 4.0 defects per 100 units.

4.3.2 Dimensional examination.- An examination shall be made to determine compliance with dimensional requirements. Any dimension that is not within the specified tolerances shall be classified a defect. The inspection level shall be S-3 with an AQL of 6.5 defects per 100 units.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:39Z
Check the source to verify that this is the current version before use.

4.3.3 Testing of the end item.- Testing shall be performed on each sample unit in accordance with 4.5.1, 4.5.2, 4.5.3, and 4.5.4. The inspection level shall be S-2 and any failure to meet the specified requirements shall be cause for rejection of the lot.

4.4 Test reports.- When required by the contract or order (see 6.2), test reports shall be furnished in duplicate showing quantitative results for all tests and analyses required by this specification.

4.5 Inspection methods.-

4.5.1 Bending radius.- The specimens for the bending test shall be of a length approximately four times the required bending radius. One end of the specimen shall be fastened tangentially to the surface of a cylindrical form so as to hang vertically, in a plane normal to the axis of the form, and the weight specified in table I shall be attached to the lower end. The radius of the form shall be as specified in table I. The form shall be slowly rotated 225 degrees to the position shown on figure 2. Failure of the conduit to come within 1/8 inch of the form, at any point along its curvature between the point of attachment and a point diametrically opposite shall constitute cause for rejection.

4.5.2 Watertightness.- Specimens for this test shall be taken from the flexed portion of specimens previously subjected to the bending radius test and shall be of a length approximately 15 times the nominal inside diameter. After removing the shielding braid, both ends shall be taped with friction or masking tape over a distance equal to the nominal inside diameter. Corks shall be fitted snugly in each end and the ends shall then be dipped in molten paraffin or wax as far back as the inner edge of the tape to insure watertightness.

4.5.2.1 Specimens prepared as noted above and in an axially straight position, shall be submerged horizontally in water so that the top of the conduit is at least 2 inches below the surface of the water. After submersion as specified in 3.2.2.3 the specimen shall be quickly and carefully removed. The contents (admitted water) shall be drained into a suitable receptacle, and weighed for calculation of amount of leakage. For the purpose of this test, the interior volume of the specimen shall be taken as the volume of a cylinder with a diameter equal to the minimum inside diameter permitted for the conduit under test, and with a length equal to the extended length of the specimen.

4.5.3 Yield load of tubing.- Specimens for this test shall be at least 7½ inches long and the shielding braid shall be removed. Gripping of the specimen in the testing machine shall be accomplished by means of plugs inserted at least 1½ inches with clamps over the plugged portion or by means of suitable jaws which do not deform the tubing. The testing machine shall be operated so that the movable head travels at a rate not exceeding 0.022 inch per minute. The drop of the beam of the testing machine, and subsequent failure of the specimen to sustain increased loading, shall be taken as the indication of yield. The final value of the yield load shall be the average of not less than two tests. Failure to meet minimum yield load requirements specified in table I shall be cause for rejection of the lot.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:39Z
Check the source to verify that this is the current version before use.

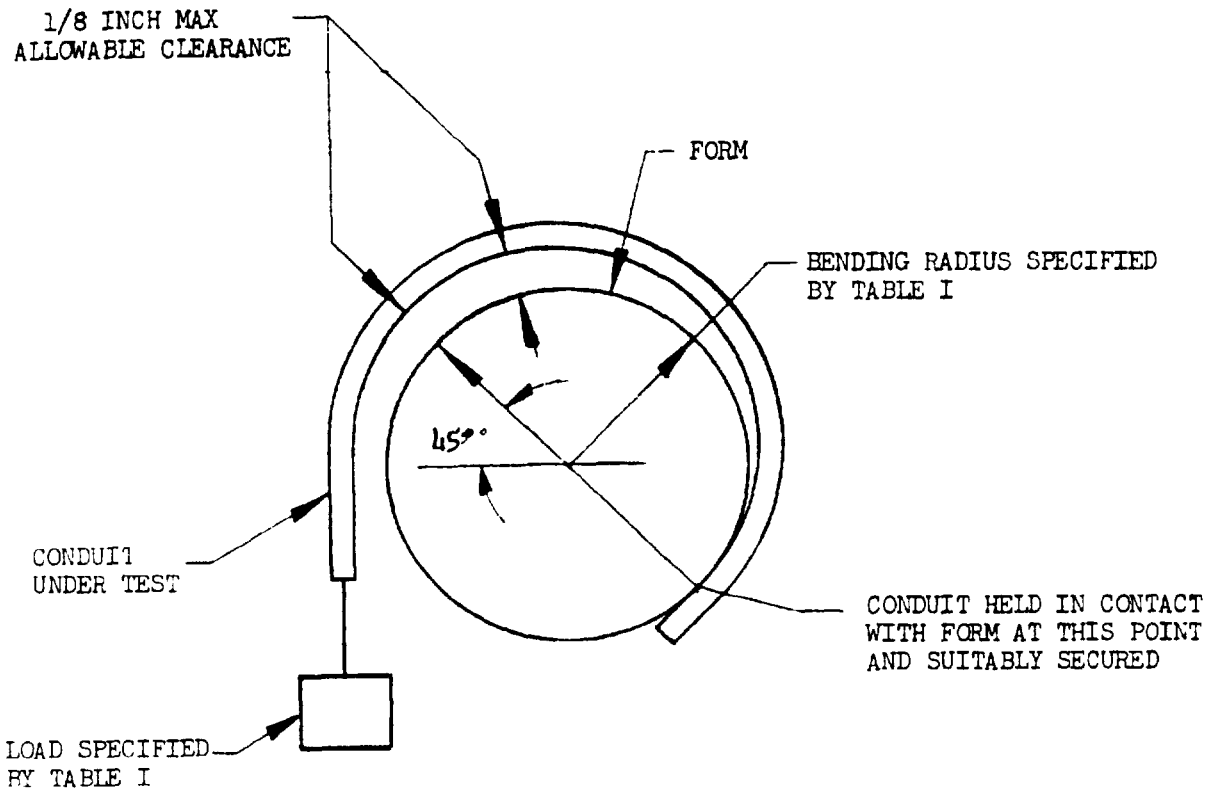



FIGURE 2. Bending radius test setup (diagrammatic)



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:39Z
Check the source to verify that this is the current version before use.

4.5.4 Braid coverage.- The coverage of the wire braid shall be determined by the following formula:

$$\text{Percent coverage} = 100\,(P + P\,(1-P))$$

Where:

$$P = \frac{AB}{C \cos \alpha}$$

A = ½ total number of wires = total number of parallel wires

B = Minimum allowable diameter of the braid wire

$$C = \text{Pitch} = \frac{A}{\text{Parallel wires per inch}}$$

$\alpha$ = Pitch angle = arc Tan C/D

D = (Max. allowable OD of conduit -2B) x 3.14

Note: The definitions contained in this specification shall apply in the use of the above formula (see 6.3).

4.6 Preparation for delivery examination.- An examination shall be made to determine compliance with the packaging, packing, and marking requirements contained in section 5. The sample unit shall be one shipping container and the lot size shall be the number of containers offered at one time. Any area of nonconformance with specified requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0 defects per 100 units.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Cleaning.- Conduit shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying.- The drying procedure used shall not be injurious to the item.

5.1.1.3 Preservation application.- None required.

5.1.1.4 Unit packaging.- Each length of conduit shall be furnished in coils of sufficient diameter to prevent excessive bending of the conduit. Each coil shall be securely tied in three places equidistantly spaced around the circumference of the coil with jute rope, filament tape or metal ties. When metal ties are used, protectors shall be applied under all ties.

5.1.2 Level C.- Each length of conduit shall be preserved and packaged in a manner that will provide adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity for immediate use. This package may conform to the supplier's commercial practice for retail distribution when such meets the requirements of this level.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:39Z
Check the source to verify that this is the current version before use.

5.2 Packing.- Packing shall be level A or C, as specified (see 6.2).

5.2.1 Level A.- Items packaged as described in paragraph 5.1.1.4 require no further packing.

5.2.2 Level C.- The packaged items shall be packed in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. This pack shall conform to the applicable carrier rules and regulations and may be the supplier's commercial practice when such conforms to the requirements of this level.

5.3 Marking.- In addition to any special marking required by the contract or order, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The flexible shielded conduit is intended for use in aircraft electrical systems.

6.2 Ordering data.- Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Nominal size of conduit (see table I).
(d) Length required. Conduit will be purchased on a length basis, the unit of length being 1 foot.
(e) Report requirements (see 4.4).
(f) Levels of preservation, packaging, and packing required (see 5.1 and 5.2).

6.3 Definitions.- (See figure 1.)

6.3.1 Total number of wires.- The total number of wires in any section taken normal to the axis of the conduit.

6.3.2 Number of parallel wires per inch.- The total number of parallel wires in one direction (to the nearest whole wire) counted along one inch of conduit length, parallel to the axis, with the conduit straight and fully extended.

6.3.3 Pick.- Any single group of parallel individual wires which are woven as a single strand.

6.3.4 Percentage of coverage.- That percentage of the surface of a cylinder, with diameter equal to the maximum allowable outside diameter of the tubing, which is covered by braid wire, as calculated by the formula specified in 4.5.4.

Custodians:
Army - EL
Navy - WP
Air Force - (85)

Preparing activity:
Navy - WP

Project No. 5975-0235

Reviewer activities
Army - EL
Navy - WP
Air Force - (85)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:39Z
Check the source to verify that this is the current version before use.

**INSTRUCTIONS.** In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

*NOTE.* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

*(Fold along this line)*

*(Fold along this line)*

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

**BUSINESS REPLY MAIL**
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Air Systems Command (AIR 51122)
Washington, DC 20361

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:39Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1 DOCUMENT NUMBER | 2 DOCUMENT TITLE |
|---|---|
| | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4 TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b ADDRESS *(Street, City State ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)* _____ |

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

| 7a NAME OF SUBMITTER *(Last First MI)* – Optional | b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c MAILING ADDRESS *(Street City State, ZIP Code)* – Optional | 8 DATE OF SUBMISSION *(YYMMDD)* |
| | |

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:39Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1 DOCUMENT NUMBER | 2 DOCUMENT TITLE |
|---|---|
| | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4 TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)* _____ |

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

| 7a NAME OF SUBMITTER *(Last, First, MI)* – Optional | b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8 DATE OF SUBMISSION *(YYMMDD)* |
| | |

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:39Z
Check the source to verify that this is the current version before use.